

Meesho

1. Given an array of size n , find the number of subarrays whose max element is equal to max_val and min element is equal to min_val .
 $N \leq 1e5, a[i] \leq 1e9, \text{max_val} \leq 1e9, \text{min_val} \leq 1e9$
Ex : [1 , 3 , 2 , 4 , 3] $\text{max_val} = 4, \text{min_val} = 2$ ans – 4
2. Given two arrays with size n , cost array and profit array, and a max_budget . If we select any i , the cost of it is $\text{cost}[i]$ and on buying it will get a profit of $\text{profit}[i]$. Now, find the maximum profit considering the max_budget .
 $N \leq 36, \text{cost}[i] \leq 1e9, \text{profit}[i] \leq 1e9$
Ex: cost – [2 , 3 , 1] profit- [4 , 5 , 2] $\text{max_budget} = 4$ ans – 6
3. Given two arrays of size n , layers and energies. Given an initial energy of k . To breach any layer, it costs $\text{layer}[i]$ of energy, and once after breaching the energy reduces by $\text{layer}[i]$, after that if the remaining energy is $\geq \text{energy}[i]$, then the person can get one point. Now need to output an array, which tells the number of points a person can earn by starting at i th index with initial energy of k .
Ex - layers – [5 , 8 , 1] energy – [4 , 2 , 1] $k = 10$ ans – [1 , 2 , 1]
Explanation, lets say the person starts at 1st index, the initial energy $\geq \text{layers}[0]$ so it will breach it, so energy becomes $10 - 5 = 5$, and this remaining energy is greater than $\text{energy}[0] - 4$, so he gets one point and he will move to second index, now the energy left is 5 and the $\text{layer}[i]$ is 8, so he cant breach that layer, so his final score is 1.
4. Given a tree, determine is it a BST or not.